

2D MONTESSORI

Dynamic Website & CMS

Product Requirements & Solution Architecture Specification

Document status	Approved scope baseline
Version	1.1
Date	12 August 2026
Platform	Flutter Web + Firebase; mobile-first and app-ready
Release	Release 1 — content website and CMS
Language	English only initially

Purpose: Define the complete functional, content, data, security, architecture, and acceptance requirements required to design, build, test, deploy, and hand over a production-grade dynamic website for 2D Montessori.

DECISION SUMMARY

Decision	Approved direction
Administrator layout control	Choose, configure, reorder, duplicate, hide, and schedule approved section templates; no free-form canvas.
Languages	English only in Release 1; schema remains localization-ready.
Business scope	Public content website plus Blog/News. No admissions, payments, portal, attendance, or operational modules.

1. Executive Summary

The product is a dynamic, production-grade Montessori content platform consisting of a public website and a secured administration application. All public content—including text, images, videos, navigation, SEO metadata, page sections, and approved layout variants—must be manageable without a code deployment.

Flutter will provide the responsive public website and administration interface. Shared packages will allow the same domain, data, design-system, and feature code to be reused when Android and iOS applications are introduced. Firebase will provide identity, database, media storage, server-side automation, hosting, monitoring, and security enforcement.

1.1 Product goals

- Present the 2D Montessori brand with a fast, accessible, search-friendly experience on mobile, tablet, and desktop.
- Allow authorized staff to safely update and publish content without developer assistance.
- Preserve design quality through approved responsive section templates and validation rules.
- Keep feature modules independently replaceable and testable through contracts, dependency inversion, and package boundaries.
- Support future native applications without redesigning the content model or business layer.

1.2 Scope

In Release 1	Out of Release 1
Home, About, Montessori Way, Experiences, Programs, Gallery, Testimonials	Admissions application and applicant tracking
Blog/News listing and articles	Fees, payments, invoices, receipts
Dynamic page builder using approved templates	Parent, student, or staff operational portals
Media library for images and video references	Attendance, academic records, messaging
Navigation, footer, branding, contact details and SEO	Arbitrary drag-and-drop/free-form page design
Admin authentication, roles, audit log and publishing workflow	Multilingual content entry and translation workflow

2. Users, Roles and Permissions

Role	Purpose	Key permissions
Public visitor	Browse published content	Read published pages, posts, programs, experiences, gallery and testimonials; use approved contact links.
Content editor	Prepare content	Create/edit drafts, upload media, preview; cannot publish, manage users, or alter global security settings.
Publisher	Control publication	Editor permissions plus review, publish, unpublish, schedule, restore

Role	Purpose	Key permissions
		prior content versions.
Administrator	Manage the platform	All Publisher permissions plus users, roles, approved templates, navigation, global settings, redirects and audit access.
System service	Background processing	Server-only media processing, scheduled publication, cache invalidation and maintenance.

Authorization must be enforced in Firebase Security Rules and trusted server functions, not only by hiding interface controls. Privileged roles must be represented using Firebase Authentication custom claims. The system must deny access by default.

3. Information Architecture and Routes

Route	Page type	Required content purpose
/	System page	Brand introduction, key value proposition, featured programs/experiences, Montessori message, testimonials and primary calls to action.
/about	Managed page	Story, mission, vision, values, founders/team and differentiators.
/montessori-way	Managed page	Explain Montessori principles, prepared environment, materials, independence and developmental approach.
/experiences	Collection listing	Published experiences with category/filter and detail routes.
/experiences/:slug	Detail	One experience: narrative, media, outcomes and related content.
/programs	Collection listing	Published programs by age/stage with summaries.
/programs/:slug	Detail	One program: ages, schedule summary, goals, activities and enquiry CTA.
/gallery	Collection page	Curated albums and media, with consent-aware publishing.
/testimonials	Collection page	Approved parent/learner stories.
/news	Collection listing	Blog/news articles, featured item, categories and pagination.
/news/:slug	Detail	Article content, author, dates, share metadata and related posts.
/:slug	Managed page	Future CMS-created informational pages using approved templates.
/404	System page	Helpful not-found state and navigation.

Header navigation and footer groups are independently managed content. Each route must have a unique canonical slug. Reserved routes cannot be reused by administrators.

3.1 UI design direction — clean, Apple-inspired

The public experience must use an Apple-inspired design direction: quiet confidence, generous whitespace, strong typographic hierarchy, high-quality edge-to-edge imagery, restrained color, rounded surfaces, subtle depth, and purposeful motion. This is a visual principle only; Apple trademarks,

layouts, copy, icons, or proprietary assets must not be copied. The Montessori identity must remain warm, human, child-centred, and recognizably 2D Montessori.

Design area	Release 1 requirement
Visual hierarchy	One dominant message per viewport section; short supporting copy; clear primary action; avoid crowded card grids and competing calls to action.
Colour	Neutral white/warm-white foundations with one approved 2D Montessori accent palette. Colour combinations must meet WCAG 2.2 AA contrast.
Typography	Use one approved sans-serif family with platform-safe fallbacks. Apply fluid type tokens; body text must remain at least 16 CSS-equivalent pixels on public pages.
Spacing	Use a consistent 4/8-point spacing system. Mobile page gutters begin at 16px, increase to 24px on larger phones/tablets, and use centred max-width content on desktop.
Surfaces	Cards use restrained radius, border, and shadow tokens. Avoid heavy gradients, glass effects, decorative borders, and excessive shadows.
Imagery	Authentic, professionally cropped Montessori photography. Templates support focal point, aspect ratio, responsive derivatives, alt text, and optional mobile-specific crop.
Icons	Use one coherent accessible icon set. Icons supporting an action require a text label or accessible name; decorative icons are excluded from semantics.
Motion	Subtle 150–300ms transitions for state changes and reveals. No essential information depends on animation; respect reduced-motion preferences.
Content density	Prefer progressive disclosure, concise copy, and breathing room. Long text uses readable measure of approximately 45–75 characters per line.

3.2 Mobile-first responsive foundation

Design and implementation start at 320px width. Mobile is the baseline—not a compressed desktop layout. Enhancements are introduced only when space permits, and every component must define its mobile, tablet, and desktop composition before development.

Viewport token	Reference range	Expected composition
compact	320–599px	Single-column flow; 16px gutter; touch-first controls; mobile header and full-screen/sheet menu; media optimized for portrait or compact crop.
medium	600–1023px	One or two columns where content remains readable; 24px gutter; mobile/tablet navigation pattern; larger media and card groups.
expanded	1024–1439px	Desktop header with visible primary navigation; centred content container; multi-column templates where approved.
wide	1440px and above	Content remains constrained by max-width tokens; whitespace expands without stretching text or creating oversized controls.

- No horizontal scrolling at 320px except an explicitly approved, accessible carousel.
- Touch targets must be at least 44×44 CSS-equivalent pixels with adequate separation.
- Layout must remain usable at 200% browser zoom and with increased text size; content cannot overlap, clip, or become unreachable.

- Responsive decisions live in the design-system and template contracts, not as page-specific magic numbers.
- Both portrait and landscape orientations must be tested on representative phones and tablets.

3.3 Header and primary navigation

State	Required behaviour
Mobile default	Sticky or static compact header, based on approved prototype, containing logo/home link and a clearly labelled Menu control. Optional single primary CTA only when it fits without crowding.
Mobile menu open	Accessible full-screen panel or bottom/side sheet with Home, About, Montessori Way, Experiences, Programs, Gallery, Testimonials, and Blog/News. Body scroll is locked; focus moves into the menu.
Nested items	Use tap-to-expand accordions; do not rely on hover. Expansion state and current page are visually and semantically communicated.
Close/return	Menu closes through Close control, Escape key, route selection, and browser navigation. Focus returns to the Menu control when dismissed.
Desktop	Logo at start, visible primary links, and optional CTA at end. Active route is clear. Overflow never silently hides links; use an approved More pattern if content exceeds available width.
Scroll behaviour	If the header changes on scroll, the transition must not cause layout shift. Header remains discoverable and must not obscure anchored headings.
Keyboard/screen reader	Logical tab order, visible focus, semantic navigation landmark, accurate expanded/current states, and no keyboard trap.

3.4 Secondary navigation and wayfinding

- Footer navigation is grouped with descriptive headings and includes contact details, social links, privacy/terms placeholders, and copyright.
- Breadcrumbs are required on deep detail pages where they improve orientation; they collapse cleanly on mobile and expose structured data where applicable.
- Collection filters use mobile-friendly chips or a filter sheet and remain keyboard accessible. The selected state and result count are announced.
- Back actions must never replace browser history semantics. Deep links and refreshes must preserve the correct route and content.
- Anchor links account for sticky-header offset, use unique labels, and move keyboard focus appropriately when needed.

3.5 Public UI component catalogue

Component	Required variants / states
App header and menu	compact/expanded; default, scrolled, menu open; current route; optional CTA
Hero	image/video/neutral; compact/full; left/centre alignment; one or two CTAs; mobile crop
Buttons and links	primary, secondary, text, icon; default, hover, focus, pressed, disabled, loading

Component	Required variants / states
Content cards	program, experience, article, gallery, testimonial; image/no image; loading and fallback
Section heading	eyebrow, title, supporting text, optional action; compact and standard spacing
Media	responsive image, video poster/player, gallery tile, lightbox; loading, error, consent-withdrawn fallback
Carousel	manual controls, indicators, swipe and keyboard; pause control if auto-advance is enabled
Accordion	closed/open, keyboard control, semantic heading/button relationship
Feedback	skeleton, progress, empty, inline error, page error, offline/retry and success states
Footer	compact/expanded groups, legal row, social/contact links

3.6 UI administration and governance

Administrators may select approved theme, spacing, alignment, media ratio, and component variants exposed by each template. They may not enter arbitrary colours, fonts, CSS, dimensions, animation values, or layout coordinates. Brand and design tokens are maintained centrally by an Administrator and changes require preview before publication.

Global UI setting	Field / control
Brand assets	primaryLogoMediaId, inverseLogoMediaId, faviconMediaId, defaultShareImageId
Colour scheme	approvedPaletteKey; light/dark/brand semantic tokens only
Typography	approvedFontKey and fixed type-scale preset
Shape and elevation	approved radius/elevation preset; no per-section numeric entry
Layout density	comfortable standard; template-defined spacing tokens
Header	headerVariant, stickyBehaviour, primaryCta, announcement visibility
Navigation	ordered items, label, destination, children, visibility, open-in-new-tab rules
Footer	ordered groups, contact block, social links, legal links, copyright text
Motion	standard reduced; system reduced-motion always overrides

3.7 UI quality acceptance

- Approved Figma or equivalent prototypes cover Home, one collection page, one detail page, mobile menu, Gallery, Blog/News, and CMS page editor at compact and expanded widths before implementation.
- Every component includes content, loading, empty, error, disabled, focus, and long-content states where applicable.
- Visual regression baselines cover 320px, 390px, 768px, 1024px, and 1440px widths for every section template and global shell.

- Real representative content—including long headings, missing optional images, multiple navigation items, and portrait/landscape media—must be used in responsive QA.
- No cumulative layout shift is introduced by fonts, images, menus, cookie/announcement surfaces, or asynchronously loaded sections.

4. CMS Page Builder

4.1 Administrator layout control

Administrators select from approved section templates. For each page they may add, configure, reorder, duplicate, hide, delete, preview, and schedule sections. Each template owns its allowed fields, validation, responsive behavior, accessibility behavior, and Flutter renderer. Administrators cannot position elements freely, inject scripts, or set arbitrary pixel coordinates.

4.2 Approved Release 1 section templates

Template key	Purpose	Principal configurable fields
hero	Opening message	eyebrow, heading, body, desktop/mobile media, overlay, alignment, height variant, primary/secondary CTA
rich_text	Narrative content	heading, body blocks, width variant, text alignment
image_text	Two-column story	heading, body, image, image side, ratio, caption, CTA
feature_grid	Benefits/values	heading, introduction, 2–6 feature cards, column variant
card_collection	Related records	heading, source collection, manual/automatic selection, item limit, card variant, CTA
stats	Key facts	heading, 2–4 stat items with value, label and optional note
quote	Highlighted statement	quote, attribution, role, portrait, background variant
testimonial_carousel	Social proof	heading, selected testimonials, auto-play flag, interval, card variant
media_gallery	Images/video	heading, selected album/items, grid variant, lightbox flag
video	Hosted/external video	title, provider, URL/storage media, poster, caption, transcript
cta_banner	Conversion prompt	heading, body, background, primary/secondary CTA
faq	Expandable answers	heading, ordered question/answer items, initial expansion state
spacer_divider	Controlled rhythm	size token or divider style; no arbitrary pixels
contact_block	Contact details	heading, body, phone, email, address, map link, hours, CTA

4.3 Common section fields

Field	Type / rule	Required	Description
id	UUID/string, immutable	Yes	Stable section identity used for editing, analytics, and rendering keys.

Field	Type / rule	Required	Description
templateKey	Enum from template registry	Yes	Determines schema and renderer.
templateVersion	Integer >=1	Yes	Enables safe template migrations.
name	String, 1–80	Yes	Internal CMS label; not normally public.
isVisible	Boolean	Yes	Controls public rendering without deleting content.
sortOrder	Integer >=0	Yes	Canonical order within page.
variant	Template-defined enum	Yes	Approved visual variant only.
theme	light dark brand neutral	Yes	Approved color token group.
anchorId	Slug, optional, unique in page	No	Accessible in-page link target.
content	Typed map validated by template schema	Yes	Template-specific values.
visibility	Device/audience rule object	No	Release 1: all, mobile-only, desktop-only; use sparingly.
publishWindow	startAt/endAt timestamps	No	Optional section-level schedule.
analyticsKey	String, 1–80	No	Stable reporting label without personal data.
createdAt/createdBy	Server timestamp/user ID	Yes	Audit metadata.
updatedAt/updatedBy	Server timestamp/user ID	Yes	Audit metadata.

5. Core Content Models

All dates are stored as UTC server timestamps and displayed in the configured site timezone (initially Asia/Kolkata). All references use immutable document IDs. Slugs are lowercase ASCII with hyphens and are unique within their route namespace. Rich text is stored as validated structured blocks—not executable HTML.

5.1 Page

Field	Type / constraints	Required
id	Auto ID; immutable	Yes
pageType	home standard collectionLanding system	Yes
title	String 1–120	Yes
navigationLabel	String 1–40	No
slug	String 1–120; unique; home uses special root mapping	Yes
summary	String 0–300	No
sections	Ordered section subcollection or ordered references	Yes
status	draft inReview scheduled published archived	Yes

Field	Type / constraints	Required
publishedVersionId	Version reference	Published only
scheduledAt	Future server timestamp	Scheduled only
seo	SEO object	Yes
socialShare	Social metadata object	No
createdAt/By, updatedAt/By	Server-maintained audit fields	Yes
publishedAt/By	Server-maintained publication fields	Published only

5.2 SEO object

Field	Type / constraints	Default
metaTitle	String 10–60	Page title + site name
metaDescription	String 50–160	Page summary
canonicalPath	Validated internal path	Current public route
indexing	indexFollow noIndexFollow noIndexNoFollow	indexFollow
ogTitle	String 0–95	metaTitle
ogDescription	String 0–200	metaDescription
ogImageId	Media reference; recommended 1200×630	Site default image
structuredDataType	Organization EducationalOrganization Article FAQPage none	Derived by content type

5.3 CTA and link object

Field	Type / constraints
label	String 1–40
linkType	internal external phone email anchor
target	Validated route, HTTPS URL, tel:, mailto:, or anchor
openInNewTab	Boolean; allowed for external only
accessibilityLabel	Optional string 1–100; required when visible label is ambiguous
trackingKey	Optional string 1–80

6. Collection Models

6.1 Program

Field	Type / constraints	Required
id	Auto ID	Yes
title	String 1–100	Yes
slug	Unique slug	Yes
shortDescription	String 1–240	Yes
heroMediaId	Media reference	Yes
ageMinMonths / ageMaxMonths	Integer ≥ 0 ; min $\leq$ max	Yes
displayAgeRange	String 1–60	Yes
durationText	String 0–80	No
scheduleText	String 0–160	No
learningGoals	Array of 1–12 strings, each ≤ 160	Yes
highlights	Array of 0–12 feature objects	No
sections	Ordered approved sections	Yes
enquiryCta	CTA object	No
featured	Boolean	Yes
sortOrder	Integer ≥ 0	Yes
status + publication/audit fields	Standard workflow fields	Yes

6.2 Experience

Field	Type / constraints	Required
title / slug / shortDescription	Same conventions as Program	Yes
categoryIds	1–5 category references	Yes
coverMediaId	Published media reference	Yes
occurredOn	Date; optional for evergreen content	No
location	String 0–120	No
participantAgeText	String 0–80	No
learningOutcomes	Array 0–12 strings	No
sections	Ordered approved sections	Yes

Field	Type / constraints	Required
relatedProgramIds	Array 0–6 references	No
featured / sortOrder	Boolean / integer	Yes
status + publication/audit fields	Standard workflow fields	Yes

6.3 Blog/News article

Field	Type / constraints	Required
title	String 1–120	Yes
slug	Unique news slug	Yes
excerpt	String 1–300	Yes
coverMediaId	Media reference	Yes
articleType	blog news announcement	Yes
categoryIds	1–3 category references	Yes
tagIds	0–10 tag references	No
authorId / authorDisplayName	User reference plus publication snapshot	Yes
bodyBlocks	Validated rich text blocks	Yes
readingMinutes	Server-computed integer	Yes
featured	Boolean	Yes
publishedAt	Server timestamp; supports scheduling	Published only
seo / socialShare	Standard objects	Yes
status + publication/audit fields	Standard workflow fields	Yes

6.4 Gallery album and media placement

Field	Type / constraints	Required
album.title / slug	String and unique slug	Yes
album.description	String 0–500	No
album.coverMediaId	Media reference	Yes
album.items	Ordered placement references	Yes
placement.mediaId	Published media reference	Yes
placement.caption	String 0–240	No

Field	Type / constraints	Required
placement.altTextOverride	String 0–180	No
placement.focalPoint	x/y decimal 0–1	No
placement.sortOrder	Integer >=0	Yes
placement.isVisible	Boolean	Yes
album.status + audit fields	Standard workflow fields	Yes

6.5 Testimonial

Field	Type / constraints	Required
quote	String 20–700	Yes
personName	String 1–80; may be initials/pseudonym	Yes
personRole	Parent Alumni Educator Learner Other	Yes
childNameOrClass	Optional string; publish only with consent	No
portraitMediaId	Optional media reference	No
rating	Optional integer 1–5	No
consentConfirmed	Boolean; must be true to publish	Yes
consentReference	Restricted string/document reference	Yes
featured / sortOrder	Boolean / integer	Yes
status + audit fields	Standard workflow fields	Yes

7. Media Library

Original media is private by default. Public delivery uses approved derivatives or explicitly published objects. The CMS must not expose raw storage paths or allow unrestricted executable uploads.

Field	Type / constraints	Required
id	Auto ID	Yes
kind	image video document	Yes
originalFileName	Sanitized display string	Yes
storagePath	Server-generated; immutable to clients	Yes
mimeType	Allowlisted MIME	Yes
sizeBytes	Integer; validated server-side	Yes
width / height / durationMs	Extracted metadata where applicable	Conditional
title	String 0–120	No

Field	Type / constraints	Required
altText	String 1–180 for meaningful public images	Conditional
caption	String 0–240	No
credit	String 0–120	No
tags	Array 0–20 strings	No
consentStatus	notRequired pending confirmed expired withdrawn	Yes
consentReference	Restricted reference; never publicly returned	Conditional
focalPoint	x/y decimals 0–1	No
processingStatus	uploaded processing ready failed quarantined	Yes
derivatives	Map of approved rendition metadata	Server
visibility	private public	Yes
createdAt/By, updatedAt/By	Audit fields	Yes

7.1 Media rules

- Images: JPEG, PNG, WebP, and AVIF input as supported; block SVG unless sanitized by a trusted process. Generate responsive WebP/AVIF/JPEG derivatives and thumbnails.
- Video: prefer a managed external streaming provider or Firebase Storage with explicit size limits and poster images. Release 1 CMS may store YouTube/Vimeo links; privacy-enhanced embedding is required where available.
- Require alt text before publishing meaningful images. Decorative media is explicitly marked decorative and emits empty alt text.
- Prevent publication where consent is pending, expired, or withdrawn. Withdrawal must remove public references and invalidate delivery caches.
- Strip unnecessary metadata such as GPS coordinates from public derivatives.
- Use checksum/deduplication, resumable uploads, progress, retry, and clear processing failure states.

8. Global Configuration

Area	Fields
Brand	siteName, legalName, logoLightId, logoDarkId, faviconId, primary/secondary/accent design tokens
Contact	primaryPhone, WhatsApp number, publicEmail, address lines, city, state, postal code, country, mapUrl, businessHours
Social	Instagram, Facebook, YouTube, LinkedIn and other allowlisted HTTPS profiles
Navigation	menu groups, labels, internal targets, order, enabled flag, new-tab rule
Footer	summary, link groups, contact visibility, copyright template, policy links

Area	Fields
SEO defaults	title suffix, default description, default social image, organization schema fields
Analytics	consent mode, analytics provider configuration references; secrets remain server-side
Feature flags	blogEnabled, experiencesEnabled, testimonialsEnabled, maintenanceMode and controlled rollout flags
System	siteTimezone, reserved slugs, records per page, upload limits; localizationReady=true

9. Editorial Workflow and Versioning

State	Who can enter	Permitted next states	Public effect
Draft	Editor+	In Review, Archived	No public change
In Review	Editor submits; Publisher reviews	Draft, Scheduled, Published	No public change
Scheduled	Publisher+	Draft, Published, Archived	Automatically publishes at scheduledAt
Published	Publisher+	Draft copy, Archived, Unpublished	Active immutable published snapshot
Archived	Publisher+	Draft restoration	Not public; retained per policy

- Editing published content creates a new draft; visitors continue to see the last published snapshot until the replacement is published.
- Every publish action stores a complete immutable version, change summary, publisher ID, and server timestamp.
- Preview uses a short-lived, authorized preview token or authenticated route and must never be indexed.
- Scheduled publishing is executed by a trusted scheduled function with idempotency and retry.
- Rollback republishes a selected prior version as a new current version; history is never rewritten.
- Deleting referenced content is blocked. Archival is preferred; hard deletion requires Administrator confirmation and retention checks.

9.1 Audit event fields

Field	Rule
id	Auto ID
eventType	create update submitReview approve publish unpublish archive restore login roleChange mediaConsentChange
entityType/entityId	Target resource identity
actorId / actorRole	Authenticated identity snapshot
timestamp	Server timestamp

Field	Rule
changeSummary	Human-entered for publication; generated for other events
changedFields	Safe field-name list or redacted diff; never log secrets or consent documents
requestId	Correlation ID
source	cms function migration

10. Functional Requirements

ID	Requirement	Priority
FR-01	Render only published snapshots to unauthenticated visitors.	Must
FR-02	CMS users can create, preview, validate, submit, publish, schedule, unpublish, archive, version, and restore content according to role.	Must
FR-03	Administrators can compose pages only from the approved template registry and approved variants.	Must
FR-04	Text, media, CTA links, visibility, ordering, layout variant, navigation and SEO are editable without deployment.	Must
FR-05	Provide collection management for Programs, Experiences, Gallery Albums, Testimonials, Categories, Tags and Blog/News.	Must
FR-06	Provide media upload, processing status, search, filtering, usage/reference display and safe replacement.	Must
FR-07	Validate slugs, required fields, accessibility fields, links, references, consent and SEO before publication.	Must
FR-08	Support responsive preview for mobile, tablet and desktop breakpoints.	Must
FR-09	Generate sitemap.xml, robots.txt behavior, canonical metadata and structured data from published content.	Must
FR-10	Provide not-found handling and managed permanent redirects when a published slug changes.	Must
FR-11	Allow draft autosave and warn about concurrent edits; optimistic version checks prevent silent overwrites.	Must
FR-12	Provide CMS search/filter by title, type, status, author, date, category and featured state.	Should
FR-13	Expose read-only, versioned content repositories usable by future Flutter mobile applications.	Must
FR-14	Record security-relevant and editorial audit events.	Must
FR-15	Allow approved content export/import for controlled migration and backup recovery.	Should

11. Architecture

11.1 Architectural style

Use a modular monolith at the client and Firebase boundary, organized as independently testable feature packages. “No dependencies” is interpreted as no direct feature-to-feature dependency: modules communicate through stable contracts, shared kernel types, repositories, and events. Complete zero-dependency modules are not technically viable because every Flutter feature requires framework and shared contracts; the goal is low coupling and independent replacement.

Layer/package	Responsibilities	Allowed dependencies
apps/public_web	Routing, composition root, public shell, bootstrap	Feature public APIs, design system, core contracts
apps/admin_web	Admin routing, shell, authentication gates	Feature admin APIs, design system, core contracts
features/<feature>/domain	Entities, value objects, use cases, repository interfaces	Dart only; core result/value types
features/<feature>/data	DTOs, mappers, Firebase repository implementations	Own domain interfaces, Firebase adapters
features/<feature>/presentation	Flutter screens/widgets/state	Own domain use cases, design system
core/contracts	Shared IDs, results, pagination, clocks, events	Dart only
core/design_system	Tokens, accessible primitives, responsive utilities	Flutter only; no feature imports
core/firebase_adapters	Auth, Firestore, Storage, Functions adapters	Firebase SDKs; core contracts
functions/<capability>	Trusted publish, media, scheduling, redirects, roles	Admin SDK; schema contracts; no UI code

11.2 Feature modules

Module	Owns
Identity & Access	Authentication session, user profile, claims display, authorization gates
Pages & Sections	Managed pages, template registry, ordered sections, previews
Programs	Program collection and detail content
Experiences	Experience collection, categories and related programs
Gallery & Media	Albums, placements, uploads, derivatives, consent state
Testimonials	Testimonials and consent publication rules
News	Articles, categories, tags, authors and archive
Navigation & Settings	Brand, menus, footer, contact, SEO defaults and flags
Publishing & Versions	Workflow, immutable snapshots, scheduling, rollback
Audit & Operations	Audit events, health visibility, redirect management

11.3 Firebase services

Service	Use
Firebase Authentication	Email/password or approved identity provider for CMS; MFA strongly recommended for Administrator and Publisher.
Cloud Firestore	Draft content, published snapshots, settings, references, versions, audit metadata and redirect maps.
Cloud Storage	Original media and derivatives with separate private/public paths.
Cloud Functions / Cloud Run	Publication transactions, role administration, media processing, scheduled jobs, sitemap/redirect generation, notifications and secure webhooks.
Firebase Hosting / App Hosting	Flutter web delivery, CDN, TLS, rewrites, headers and release channels.
App Check	Protect callable/backend resources from unauthorized clients where supported.
Crashlytics / Performance / Analytics	Mobile-ready observability and privacy-aware usage insights.
Cloud Logging / Monitoring	Function errors, latency, scheduled job failures, alerts and operational dashboards.

12. Firestore Logical Model

Collection/path	Purpose	Public read
users/{uid}	CMS profile and non-authoritative role display	No
pages/{pageId}	Editable page head/current workflow state	No
pages/{pageId}/sections/{sectionId}	Editable typed sections	No
pages/{pageId}/versions/{versionId}	Immutable version metadata/snapshot reference	No
publishedPages/{routeKey}	Denormalized published page snapshot	Yes
programs, experiences, articles	Editable collection records	No
publishedPrograms, publishedExperiences, publishedArticles	Query-optimized published snapshots	Yes
albums/{id}, testimonials/{id}	Editable content	No
publishedAlbums, publishedTestimonials	Consent-safe public snapshots	Yes
media/{mediaId}	Media metadata including restricted fields	No
publicMedia/{mediaId}	Sanitized public delivery metadata only	Yes
categories/{id}, tags/{id}	Taxonomy master	Published subset only
settings/{key}	Editable global configuration	No

Collection/path	Purpose	Public read
publicSettings/{key}	Sanitized published configuration	Yes
redirects/{sourceKey}	Redirect master	Resolved server-side
auditLogs/{eventId}	Append-only event history	No
publishJobs/{jobId}	Idempotent schedule/processing job state	No

Published collections are deliberate read models. They prevent draft, consent, security, and internal audit fields from leaking and reduce public query complexity. Publication must validate all references and atomically update the public snapshot, version record, status, redirects, and audit event through a trusted backend process.

12.1 Index and query expectations

- Published listing queries: status-equivalent public collection ordered by featured then sortOrder/publishedAt; category plus publishedAt; slug equality.
- CMS queries: content type plus status and updatedAt; author plus updatedAt; category plus status; title search may use normalized prefixes or an approved external search service later.
- Avoid unbounded arrays and documents approaching Firestore limits. Store high-volume sections/items in subcollections or version snapshot objects in Storage where required.
- Pagination must use cursors, never large offset scans. Query limits and maximum page sizes are configured centrally.

13. Security and Privacy Requirements

- Deny-by-default Firestore and Storage rules; public clients read only sanitized published collections and approved public derivatives.
- Only trusted backend code changes custom claims, publishes snapshots, writes append-only audit events, or marks media processing complete.
- Require recent authentication for role changes and destructive administrative actions; MFA for privileged roles.
- Validate MIME type, file signature, size, dimensions, ownership, and path on upload; quarantine failures.
- Apply Content Security Policy, HSTS, X-Content-Type-Options, Referrer-Policy and frame restrictions compatible with approved embeds.
- No secret keys, service credentials, consent documents, personal contact details, or internal notes in client bundles or public documents.
- Collect the minimum personal data. Testimonials and identifiable child media require recorded consent and a withdrawal process.
- Use separate Firebase projects for development, staging, and production, with separate credentials, buckets, indexes, rules and deployment approval.
- Rate-limit public write/callable endpoints. Release 1 should use direct mail/phone/WhatsApp links rather than an open enquiry database unless a protected form is separately approved.
- Define retention and deletion schedules for audit events, draft history, originals, consent evidence and inactive user accounts before production launch.

14. Non-Functional Requirements

Area	Acceptance target
Availability	Public site monthly target 99.9%, excluding announced maintenance and upstream outages.
Performance	At p75 on representative mobile: LCP <=2.5s, INP <=200ms, CLS <=0.1 for principal public templates; compressed responsive media and lazy loading required.
Accessibility	WCAG 2.2 AA target: keyboard navigation, visible focus, semantic landmarks, contrast, text scaling, reduced motion, captions/transcripts and meaningful alt text.
Responsive design	Supported from 320px width; defined mobile, tablet and desktop behavior for every approved template.
Browser support	Current and previous major versions of Chrome, Edge, Safari and Firefox; test current iOS Safari and Android Chrome.
SEO	Indexable rendered content, unique metadata, canonical URLs, structured data, sitemap, robots controls, redirects and meaningful status codes. Flutter implementation must prove crawlability in staging.
Scalability	Read model and CDN-first media delivery; no full collection scans; cursor pagination; idempotent background jobs.
Reliability	Atomic publication, retries with backoff, dead-letter/failed state visibility, rollback and automated backup verification.
Maintainability	Feature boundaries, linting, documented contracts, >=80% unit coverage for domain/use-case code, integration tests for repositories and rules.
Observability	Structured logs with request IDs, error reporting, latency/error dashboards, quota alerts, publish-job alerts and runbooks.
Privacy	Consent-safe media/testimonial publishing, metadata stripping, least privilege and documented retention.

15. Admin Experience Requirements

- Dashboard: content counts by status, scheduled items, recent activity, failed media/jobs, and quick-create actions.
- Content editor: autosave indicator, field-level validation, unsaved-change guard, change summary, revision comparison, preview and workflow actions.
- Page editor: section outline plus selected-template form; keyboard reorder controls in addition to pointer controls.
- Media picker: thumbnail/list views, search/filter, usage references, consent status, dimensions, alt-text completeness and safe replace.
- Publish checklist: missing fields, broken/internal links, unpublished references, SEO warnings, accessibility issues, media processing and consent blockers.
- User management: invite/disable users, set permitted roles, require recent authentication, and audit every role change.
- Operational clarity: actionable error messages, retry actions, no raw stack traces, and links to affected content.

16. Testing and Quality Gates

Test level	Required coverage
Unit	Entities, value objects, validation, use cases, permissions, scheduling and mappings.
Widget	Responsive templates, editor forms, validation states, accessibility semantics and navigation.
Repository integration	Firestore converters, queries, pagination, Storage lifecycle and emulator-backed behavior.
Security rules	Public/draft separation, role matrix, ownership, denied writes, media paths and regression tests using Emulator Suite.
Cloud functions	Idempotency, authentication, authorization, transactions, retries, scheduling, audit creation and failure states.
End-to-end	Editor creates draft → Publisher reviews → preview → publish → public render → update → rollback. Include slug redirect and consent withdrawal.
Visual regression	Every approved section template at mobile/tablet/desktop widths and each theme/variant.
Accessibility	Automated scans plus manual keyboard, focus, screen-reader landmarks, zoom and reduced-motion checks.
Performance/SEO	Lighthouse CI budgets, crawl/index staging verification, metadata, sitemap, structured data and redirect tests.
Disaster recovery	Documented and rehearsed restore from Firestore/Storage backups to a non-production project.

16.1 Definition of done

- Requirement has acceptance tests and product approval.
- Implementation respects module boundaries and contains no feature-to-feature internal imports.
- All automated tests, linting, static analysis, security rules tests and performance budgets pass.
- CMS permission and validation behavior is verified for every role.
- Responsive, accessibility, error, loading and empty states are complete.
- Telemetry contains no sensitive data and operational alerts/runbooks are updated.
- Documentation, schema/version changes, indexes and deployment scripts are committed and reviewed.

17. Release and Operations

- Environments: Development, Staging, Production. Staging uses production-equivalent rules, indexes, headers and representative non-sensitive content.
- CI validates formatting, analysis, tests, rules, functions, indexes, build size and Lighthouse thresholds. CD deploys preview channels, then staging, then production with approval.
- Use backward-compatible content/template migrations. A templateVersion identifies renderer/schema compatibility; migrations are idempotent and auditable.
- Back up Firestore and original media per recovery policy. Define target RPO of 24 hours and RTO of 4 hours for Release 1, then validate with a restore drill.

- Maintain runbooks for failed publication, stuck scheduled job, invalid role claims, broken media processing, rollback, consent withdrawal and quota exhaustion.
- Domain, DNS, TLS, Firebase ownership, billing alerts, access recovery, and break-glass administrator procedures must be documented before launch.

18. Phased Delivery Plan

Phase	Outcome
0 — Discovery and UX	Approved sitemap, content inventory, design tokens, section template catalogue, data dictionary, analytics/privacy decisions and migration plan.
1 — Foundation	Flutter workspace, modular packages, Firebase projects/emulators, authentication, claims, rules, CI/CD, design system and observability baseline.
2 — CMS Core	Pages/sections, template registry, media library, settings, navigation, roles, preview, validation and audit.
3 — Public Website	All public routes, Programs, Experiences, Gallery, Testimonials, Blog/News, SEO, accessibility and responsive behavior.
4 — Publishing & Hardening	Versioning, scheduling, rollback, redirects, performance, security review, backups, recovery drill and content migration.
5 — Launch & Handover	Production launch, editor training, operating handbook, source/schema documentation, warranty/support and backlog.

19. Acceptance Criteria by Epic

Epic	Release acceptance
Dynamic pages	An Administrator can assemble every required page using only approved templates; changes remain draft until published and render correctly at all supported breakpoints.
UI and navigation	The approved Apple-inspired design system is applied consistently; mobile navigation is fully operable by touch, keyboard and screen reader from 320px; layouts pass defined breakpoint, zoom, long-content and visual-regression tests.
Publishing	A Publisher can review, schedule, publish, unpublish and restore; public readers never receive draft or internal fields.
Collections	Programs, Experiences, Albums, Testimonials and Articles support required fields, ordering, taxonomy, preview, publication and public routes.
Media	Uploads are validated and processed; responsive derivatives, alt text, focal point and usage are visible; consent blockers prevent publication.
SEO	Every public route emits correct title, description, canonical, social metadata and applicable structured data; sitemap, redirects and 404 behavior pass tests.
Security	Emulator security tests prove the role matrix and public/draft boundary; role changes and publish operations are server-enforced and audited.
Mobile readiness	Domain/use-case/repository contracts and design-system packages compile independently of web shells and can be consumed by a Flutter mobile app without schema redesign.

Epic	Release acceptance
Operations	Staging/production deploys are repeatable; alerts, backups, restore test, rollback and support runbooks are accepted.

20. Assumptions, Constraints and Deferred Decisions

20.1 Confirmed assumptions

- English is the only authoring and public language in Release 1, but user-visible strings and schema design must not prevent later localization.
- Administrators use approved templates; free-form layouts and custom code are prohibited.
- Blog/News is included. “Content website only” excludes transactional and school-operation features.
- Contact actions use approved external channels in Release 1 unless a separately secured enquiry form is approved.
- The existing website supplies content direction; the new system is a rebuilt dynamic platform, not a pixel-identical copy.

20.2 Decisions to complete during discovery (do not block this baseline)

Decision	Recommended default
CMS sign-in method	Email/password with MFA for Publisher/Admin; consider Google Workspace SSO if accounts are centrally managed.
Video delivery	YouTube/Vimeo privacy-enhanced embeds for most content; Storage only for short, optimized assets after cost/performance review.
Search	Filtered collections and title-prefix CMS search in Release 1; external full-text search only when content volume proves the need.
Analytics and cookies	Privacy-minimal analytics with consent mode; finalize policy and banner based on enabled providers.
Content migration	Manual curated migration with media optimization, alt text, consent confirmation and redirect mapping.
Retention	Approve exact durations with legal/privacy owner before launch.

Appendix A — Validation Summary

Data type	Baseline rule
Short label	Trimmed; 1–40 characters
Title	Trimmed; 1–120 characters depending on entity
Slug	Lowercase ASCII letters/numbers/hyphens; no leading/trailing or duplicate hyphens; unique in namespace

Data type	Baseline rule
Summary/excerpt	Plain text; entity-specific maximum; no embedded markup
Rich text	Allowlisted structured blocks and marks; sanitized links; no scripts, iframes, inline styles or arbitrary HTML
URL	HTTPS except validated phone/email/anchor schemes; external host allowlist for embeds
Image alt text	1–180 characters when meaningful; explicitly decorative otherwise
Sort order	Non-negative integer; normalized transactionally after reorder
References	Must exist and, at publish time, resolve to publishable/published records
Timestamps	Trusted server timestamps; schedule validated in site timezone then stored UTC
Consent	Confirmed and not expired/withdrawn at publication and delivery time
File	Allowlisted MIME and signature; configured size/dimension limits; sanitized filename

Appendix B — Recommended Repository Structure

- apps/public_web — public Flutter web composition root
- apps/admin_web — secured CMS composition root
- packages/core_contracts — IDs, result/error, pagination, clocks and events
- packages/design_system — tokens and accessible responsive primitives
- packages/firebase_adapters — shared Firebase implementation adapters
- packages/feature_pages — domain/data/presentation plus public API
- packages/feature_programs, feature_experiences, feature_gallery, feature_testimonials, feature_news
- packages/feature_media, feature_identity, feature_settings, feature_publishing, feature_audit
- functions/src/publishing, media, scheduling, identity, redirects, maintenance
- firebase/firestore.rules, storage.rules, indexes, emulator fixtures
- tool/migrations — versioned, idempotent content/template migrations
- docs/architecture — ADRs, schema, permissions, runbooks and release procedures

Appendix C — Traceability Baseline

Every implementation story must reference one or more requirement IDs or a named model/section in this document. Any scope addition—especially forms collecting personal data, admissions, payments, portals, multilingual authoring, arbitrary layouts, or custom embeds—requires a documented change request with data, security, cost, testing, and operational impact assessment.